1. What are the header fields to identify a UDP socket and a TCP socket?
2. Please describe the mechanism of packet error detection using checksum.
3. Please calculate the checksum of the following two 16-bit binary integers: 1110011001100110 & 1101010101010101.
4. Please list the network applications of UDP.
5. Please describe all necessary mechanisms to recover the packets with bit errors.
6. Please explain why one-bit sequence number is enough for a stop-and-wait protocol.
7. What is the function of sequence numbers in feedback segments?
8. Please describe all necessary mechanisms to recover packet loss.
9. Please describe the problems caused by improper timeout values.
10. The performance of the stop-and-wait protocol is poor and can be improved by a pipelining protocol. Please describe the idea of the pipelining protocol.
11. Please describe the difference between Go-Back-N and Selective Repeat for packet retransmission, receiver window (buffer) size, and number of timers.
12. How many sequence numbers are required for selective repeat? (window size=W)
13. Please introduce the basic idea of calculating TCP timeout value.
14. Please describe the procedure of TCP delayed ACK in TCP receivers.
15. Please describe the idea of TCP Fast Retransmit.
16. TCP flow control can avoid receiver buffer overflows. Please describe its procedure.
17. Please describe the difference of reliable data transfer between TCP and Go-Back-N.
18. Please explain the values specified in fields of sequence number and acknowledge number in a TCP header.
19. TCP uses three-way handshake for connection setup. Why is two-way handshake not used?
20. Please describe the four-way handshake of closing a TCP connection.
21. Please describe how TCP slowstart phase adjusts CWND value.

22. Please describe the idea of AIMD (additive increase; multiplicative decrease) in TCP for adjusting CWND.